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Doubly-Unlinked Regression for Dependent Data

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Abstract

Modern spatial and temporal datasets often violate the standard assumption that covariates, responses, and their underlying domain are correctly aligned. In practice, correspondence may fail in two ways: the covariate–response link may be broken when exposures and outcomes are observed without their correct pairing, and the response–domain link may be broken when responses are detached from the spatial or temporal structure inducing their dependence. While existing work studies unlinked regression with a single missing correspondence under independent observations, the simultaneous loss of both correspondences in dependent data has not been systematically studied. We introduce the doubly-unlinked regression model, in which covariates and latent domain processes are observed through two unknown permutations that break the covariate–response and response–domain links. We characterize signal-to-noise regimes governing recovery of the regression parameter and the latent permutations, and show that consistent estimation of the regression effect can be achieved under strictly weaker signal conditions than those required for exact permutation recovery. Consequently, valid inference on the regression signal can remain feasible even when the underlying correspondences cannot be recovered. We develop REPAIR, a variational Bayes method for joint estimation of the regression parameter, latent permutations, and parameters governing the covariance structure. Simulations and an application to a real-world dataset illustrate the empirical performance of the proposed approach.

Key words: Shuffled regression, Dependent data, Spatial data, Exponential covariance, Permutation inference, Variational Bayes

1. Introduction

Modern spatial and temporal datasets are typically analyzed under the assumption that covariates, responses, and their underlying spatial or temporal domain are correctly aligned. In practice, however, these correspondences may be partially or completely missing. Two distinct failures can occur. The first is a broken covariate–response link: in survey-based epidemiological studies, exposure records, and health outcomes are often drawn from separate registries and linked imperfectly, leaving the covariate–response pairing uncertain (Lahiri and Larsen, 2005); in genomics, sample mix-ups during high-throughput sequencing attaches gene expression profiles to the wrong phenotypic outcomes (Westra et al., 2011). The second is a broken response–domain link: dissociation-based transcriptomics protocols lyse individual cells from tissue to sequence their transcriptomes, discarding spatial positional identity in the process (Nitzan et al., 2019); analogous ordering failures arise in pseudo-time inference (Gu et al., 2022), seriation in archaeology (Banning, 2020), and spike sorting in neural arrays (Rossant et al., 2016). Both failures also arise deliberately when data are released under geoprivacy constraints that obscure spatial locations or covariate pairings to protect individual identities (Lin, 2023; Cassa et al., 2008). These settings raise a fundamental question about whether recovering the domain correspondence is necessary for inferring the underlying regression signal, or whether valid inference can proceed even when the permutations linking covariates, responses, and domain structure remain unknown. In this paper, we develop theory and methods to address both settings.

Let $\mathbf{X} = (X(s_1), \dots, X(s_n))^\top$ denote the covariate vector, let $\mathbf{Y} = (Y(s_1), \dots, Y(s_n))^\top$ denote the response vector, and let $\mathbf{W} = (W(s_1), \dots, W(s_n))^\top$

denote a latent stochastic process at sampled domain points $s_1, \dots, s_n$. We consider the joint loss of correspondence through the data generating process (DGP)

$$\mathbf{Y} = \mathbf{\Pi}_X \mathbf{X} \beta + \mathbf{\Pi}_S \mathbf{W} + \boldsymbol{\epsilon}, \quad (1)$$

where $\mathbf{\Pi}_X$ and $\mathbf{\Pi}_S$ are unknown permutation matrices. The permutation $\mathbf{\Pi}_X$ breaks the covariate–response link while $\mathbf{\Pi}_S$ breaks the response–domain link through which dependence is induced. The joint problem is not a superposition of two known single-link problems. Recovery of $\mathbf{\Pi}_X$ alone is polynomial-time tractable in iid settings (Zhang and Li, 2020), but recovery of $\mathbf{\Pi}_S$ leads to a quadratic assignment problem that is NP-hard (Loiola et al., 2007); their coupling through β and the structure of the process covariance make the combined problem harder. More fundamentally, the latent dependence alters the information geometry in a way that creates distinct recovery regimes for β and for $(\mathbf{\Pi}_X, \mathbf{\Pi}_S)$, a phenomenon not explored in the existing iid shuffled regression literature.

The theory of unlinked regression under iid observations is now well-developed. Foundational work established identifiability and recovery limits: Unnikrishnan et al. (2015) showed that the permutation is generically identifiable from unlabeled measurements, and Pananjady et al. (2017) characterised a sharp signal-to-noise threshold below which permutation recovery is information-theoretically impossible regardless of computational budget. Subsequent work derived statistically and computationally efficient joint estimators (Zhang and Li, 2020), polynomial-time convex relaxations (Slawski and Ben-David, 2019; Pananjady et al., 2018), and extensions to spectral, graph-based, and optimal transport formulations (Hsu et al., 2017; Liu and Scaglione, 2025; Liu et al., 2024; Beuthner et al., 2021; Slawski et al., 2024; Wang et al., 2023; Durot and Mukherjee, 2024; Fabrizi et al., 2025). All of this

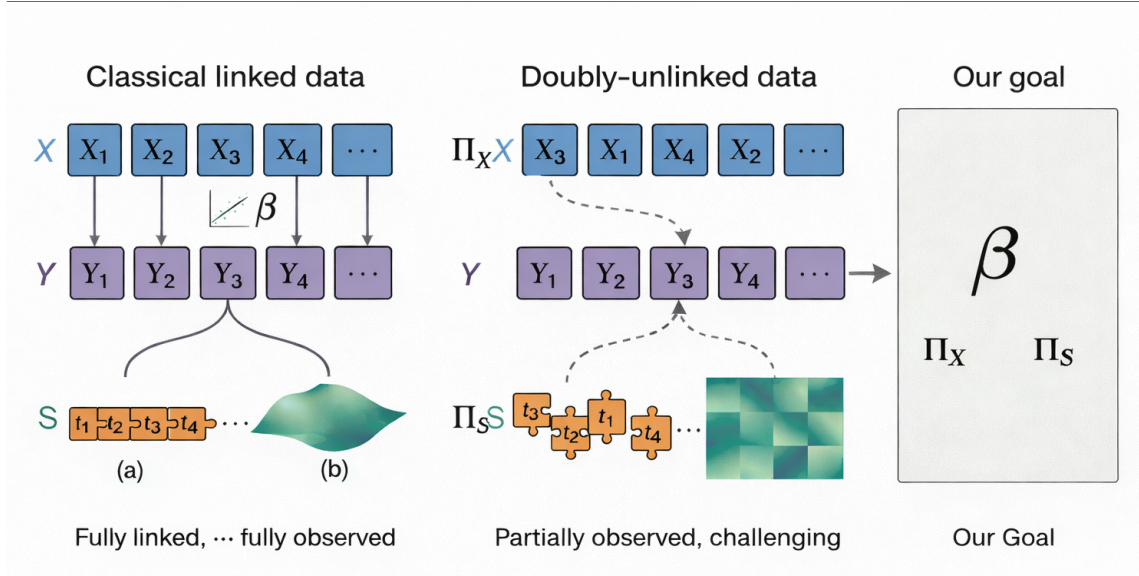


Fig. 1: Schematic illustration of the doubly-unlinked regression setting. The left panel shows the classical linked-data setting with aligned exposures, outcomes, and domain variables and regression effect β . The middle panel depicts the doubly-unlinked setting, where exposures and domain variables are observed after unknown permutations. The primary inferential target is β , while recovery of π_X and π_S is secondary.

work assumes iid observations and a single broken link; none addresses the spatial or temporal dependence that characterises the applications motivating this paper. The doubly-unlinked model studied here involves both failures simultaneously under structured dependence, and produces a qualitatively new phenomenon: the threshold governing permutation recovery and the condition for consistent β estimation are strictly different, a separation with no counterpart in the existing iid shuffled regression literature.

This paper makes three contributions. First, we introduce the doubly-unlinked regression setup (1) as the first systematic framework for regression under simultaneous covariate–response and response–domain correspondence failures in

dependent data, and provide a theoretical characterisation of the signal-to-noise conditions governing permutation and regression recovery. The central finding is a separation: the SNR condition sufficient for consistent estimation of β is strictly weaker than the condition sufficient for exact permutation recovery, so that valid regression inference remains feasible in signal regimes where permutation recovery is not guaranteed. This separation has a natural implication for spatial data release. Most spatial regression models already adopt a Gaussian process prior, under which individual locations are treated as samples from a smooth latent surface rather than as fixed identities. The inferential target is then the population-level exposure effect β , not individual-level spatial attribution. Our results show that a structured scrambling of both covariate assignments and spatial coordinates can protect individual-level correspondence while leaving β consistently estimable, pointing to a class of privacy mechanisms distinct from spatial aggregation or noise addition. Second, we develop REPAIR (**RE**gression with **P**ermutation **A**lignment via Variational **I**nfe**R**ence), a variational Bayes algorithm for joint inference on β , the two latent permutations matrices, and the covariance parameters of the latent process, built on a block-structured permutation model that captures localized scrambling. Third, simulation studies and an applied analysis on the Meuse soil dataset confirm that REPAIR recovers β accurately across a range of settings, with near-oracle performance at high SNR; the exposure permutation π_X is reliably recovered across settings, while recovery of the spatial permutation π_S improves as the block structure becomes finer.

The remainder of the paper is organized as follows. Section 2 introduces the DGP and formal assumptions. Section 3 presents the main theoretical results. Section 4

develops REPAIR. Section 5 investigates finite-sample performance, Section 6 presents the applied analysis, and Section 7 concludes.

2. Background and Problem Setup

Notation. We use the standard order notation $\mathcal{O}(\cdot)$ and $\Omega(\cdot)$ throughout. For two nonnegative sequences a_n and b_n , we write $a_n = \mathcal{O}(b_n)$ if there exist constants $C > 0$ and N such that $a_n \leq Cb_n$ for all $n \geq N$, and $a_n = \Omega(b_n)$ if there exist constants $c > 0$ and N such that $a_n \geq cb_n$ for all $n \geq N$. For a vector $\mathbf{x} \in \mathbb{R}^d$, $\|\mathbf{x}\|_2$ denotes the Euclidean norm. For a matrix $\mathbf{A}$, $\|\mathbf{A}\|_2$ denotes the spectral norm unless stated otherwise.

Formal Problem Setup. We consider the DGP (1) over a latent domain $\mathcal{D} \subset \mathbb{R}^k$. Let $\{s_i\}_{i=1}^n \subset \mathcal{D}$ denote observed domain points, and let $\mathbf{W}$ be a mean-zero latent process with covariance matrix Σ^* . It represents structured dependence induced by the domain $\mathcal{D}$. Such dependence is ubiquitous in time-series analysis, spatial statistics, longitudinal studies, sensor networks, and manifold-based biological data. Even if individual-level correspondences are scrambled, the covariance structure induced by $W(s_i)$ reflects the geometry or ordering of the underlying domain. Our primary inferential objective is estimation of β , with recovery of the permutations treated as a secondary objective when feasible. We begin by stating the assumptions imposed on the DGP in (1). Additional structural restrictions used to make inference tractable will be introduced subsequently.

Assumption 1 (Data Generation Assumption) *We assume that the data generation process satisfies:*

- The exposure vector $\mathbf{X} \sim \mathcal{N}_n(\mathbf{0}, \sigma_X^2 \mathbf{I}_n)$.

- $W(s_i)$ is a realization from $\mathcal{GP}(0, C(\cdot, \cdot))$ where the covariance function $C(\cdot, \cdot)$ belongs to a known isotropic class indexed by parameters $\mathbf{\Gamma} = (\gamma_1, \dots, \gamma_c)^\top$.
- $\mathbf{X}$ is independent of $\mathbf{W}$, i.e., there is no latent endogeneity.
- The random error vector $\boldsymbol{\epsilon} \sim \mathcal{N}_n(\mathbf{0}, \tau^2 \mathbf{I}_n)$ is independent of $\mathbf{X}$ and $\mathbf{W}$.

Under Assumption 1, the random effects vector observed at the n domain indices $\mathbf{S} = (s_1, \dots, s_n)^\top$ satisfies $\mathbf{W} \sim \mathcal{N}_n(\mathbf{0}, \Sigma^*)$ where $\Sigma^* = \sigma^2 \mathbf{R}_{\boldsymbol{\theta}^*}$ and $\mathbf{R}_{\boldsymbol{\theta}^*}$ is the corresponding correlation matrix induced by $C(\cdot, \cdot)$ evaluated at $\mathbf{S}$, parametrised by $\mathbf{\Gamma} = (\sigma^2, \boldsymbol{\theta}^*)$. Marginalizing over $\mathbf{W}$ yields $\mathbf{Y}|\mathbf{X} \sim \mathcal{N}_n(\Pi_X \mathbf{X} \beta, \Pi_S (\Sigma^* + \tau^2 \mathbf{I}_n) \Pi_S^\top)$. Defining $\Sigma = \Sigma^* + \tau^2 \mathbf{I}_n$, the conditional covariance of $\mathbf{Y}$ is $\Pi_S \Sigma \Pi_S^\top$.

Because the latent process $\mathbf{W}$ is not observed, direct inference on Π_S is generally infeasible. For theoretical purposes, it is convenient to reparameterize (Π_S, Π_X) through the one-to-one transformation $(\Pi_S, \Pi_X) \mapsto (\Pi_1, \Pi_2)$ defined by $\Pi_1 = \Pi_S^\top \Pi_X$ and $\Pi_2 = \Pi_S^\top$. We emphasize that this reparameterization is introduced only to facilitate analysis; the methodological development will be presented in the original parametrization. Left-multiplying (1) by Π_2 gives $\Pi_2 \mathbf{Y} = \Pi_1 \mathbf{X} \beta + \mathbf{W}^*$, where $\mathbf{W}^* \sim \mathcal{N}_n(\mathbf{0}, \Sigma)$. This representation recasts the problem as a Generalized Least Squares (GLS) version of shuffled regression under correlated errors. Assuming that Σ is known, define $\boldsymbol{\Theta} := (\Pi_1, \Pi_2, \beta)$ and consider the GLS loss function

$$\begin{aligned} \mathcal{L}(\boldsymbol{\Theta}) &:= \left\| \tilde{\mathbf{Y}}_{\Pi_2} - \tilde{\mathbf{X}}_{\Pi_1} \beta \right\|_2^2 \\ &= \underbrace{\left\| \mathbf{P}_{\Pi_1, X}^\perp \tilde{\mathbf{Y}}_{\Pi_2} \right\|_2^2}_{\mathcal{L}_1(\boldsymbol{\Theta})} + \underbrace{\left\| \mathbf{P}_{\Pi_1, X} \tilde{\mathbf{Y}}_{\Pi_2} - \tilde{\mathbf{X}}_{\Pi_1} \beta \right\|_2^2}_{\mathcal{L}_2(\boldsymbol{\Theta})}, \end{aligned} \tag{2}$$

where $\widetilde{\mathbf{X}}_{\mathbf{\Pi}_1} := \Sigma^{-1/2} \mathbf{\Pi}_1 \mathbf{X}$, $\widetilde{\mathbf{Y}}_{\mathbf{\Pi}_2} := \Sigma^{-1/2} \mathbf{\Pi}_2 \mathbf{Y}$, and $\mathbf{P}_{\mathbf{\Pi}_1, X}$ is the projection matrix onto the column space of $\widetilde{\mathbf{X}}_{\mathbf{\Pi}_1}$ with $\mathbf{P}_{\mathbf{\Pi}_1, X}^\perp := \mathbf{I}_n - \mathbf{P}_{\mathbf{\Pi}_1, X}$. We need to minimize the loss function (2) in order to estimate Θ .

Block Permutation Structure. The fully unrestricted permutation model allows $(n!)^2$ possible pairs $(\mathbf{\Pi}_1, \mathbf{\Pi}_2)$, rendering both theoretical characterization and computation intractable for even moderate n . Moreover, in many practical settings, scrambling mechanisms are not globally arbitrary but instead exhibit localized structure. To reflect this and to obtain a statistically meaningful intermediate regime between complete correspondence and fully adversarial permutations, we impose a structured restriction on $\mathbf{\Pi}_1$ and $\mathbf{\Pi}_2$.

For n sampled domain indices, we partition them into B blocks of size K , so that $n = KB$. Within each block, exposure values and domain indices may be permuted, but no permutation occurs across blocks. This restriction models localized scrambling mechanisms such as batching, windowed anonymization, temporal shuffling within short time segments, or partial reordering within contiguous domain neighborhoods. In such settings, coarse-scale ordering across blocks is preserved, while fine-scale correspondences within blocks are obscured.

From a statistical perspective, the block structure creates an intermediate inferential regime: the latent dependence induced by $W(s_i)$ continues to operate across blocks, preserving large-scale covariance structure, while local permutations disrupt short-range alignment between $\mathbf{X}$ and $\mathbf{Y}$. This separation allows us to study whether regression recovery can exploit global dependence even when local correspondences are scrambled. The block model, therefore, isolates the interaction between dependence and permutation noise without requiring recovery over the full combinatorial space.

We further assume that, for the two $K \times K$ permutation matrices $\boldsymbol{\pi}_1 := \boldsymbol{\pi}_X^\top \boldsymbol{\pi}_S$ and $\boldsymbol{\pi}_2 := \boldsymbol{\pi}_S^\top$, the corresponding full permutation matrices of size $n = KB$ satisfy $\boldsymbol{\Pi}_u = \text{bdiag}(\boldsymbol{\pi}_u)$ for $u = 1, 2$ where $\text{bdiag}(\mathbf{A})$ denotes block diagonal matrix with the same $\mathbf{A}$ repeated. Thus, the same within-block permutation is repeated across all B blocks. This assumption is convenient both analytically and computationally, as it imposes a simple structure while still allowing for local scrambling within each block. It is also natural in settings where observations are perturbed according to a common mechanism across blocks; for example, when data are masked prior to release, the same permutation rule may be applied repeatedly within each block. Under this specification, there is a correspondence between $\boldsymbol{\Pi}_X$ and $\boldsymbol{\Pi}_S$ with the corresponding block matrices $\boldsymbol{\pi}_X$ and $\boldsymbol{\pi}_S$. Similarly any estimate $\hat{\boldsymbol{\Pi}}_u$ can be represented as $\text{bdiag}(\hat{\boldsymbol{\pi}}_u)$. For Hamming distance d_H between the identity matrix $\mathbf{I}_K$ and $\boldsymbol{\pi}_u$ equal to $d_H(\mathbf{I}_K, \boldsymbol{\pi}_u) = k_u$, we have $d_H(\mathbf{I}_n, \boldsymbol{\Pi}_u) = h_u = k_u B$, and analogous definitions apply to $\hat{\boldsymbol{\pi}}_u$ and $\hat{\boldsymbol{\Pi}}_u$ with corresponding $\hat{k}_u$. We will use these notations throughout.

Under Assumption 1, minimizing the GLS loss (2) over $\boldsymbol{\Theta} = (\boldsymbol{\Pi}_1, \boldsymbol{\Pi}_2, \beta)$ within the block permutation class coincides with Maximum Likelihood Estimation (MLE). We therefore study the consistency of the maximum likelihood estimators of β , $\boldsymbol{\Pi}_1$, and $\boldsymbol{\Pi}_2$ under the block model and the assumed DGP. In particular, we ask whether consistent estimation of β requires exact recovery of the permutations, or whether the latent dependence structure induced by W permits regression recovery under weaker signal-to-noise conditions. The next section establishes theoretical results characterizing the regimes governing permutation and regression consistency.

3. Theoretical Guarantees for MLE

Signal-to-Noise Ratio. The recoverability of the regression parameter β and the permutation matrices (Π_X, Π_S) under DGP (1) depends on the relative strength of the linear signal compared to the latent-domain variability. To quantify this balance, we define the Signal-to-Noise Ratio (SNR) as

$$\text{SNR} := \frac{\beta^2}{\lambda_{\max}(\Sigma)\kappa(\Sigma)},$$

where $\lambda_{\max}(\Sigma)$ is the largest eigenvalue of $\Sigma = \sigma^2 \mathbf{R}_{\theta^*} + \tau^2 \mathbf{I}_n$ and $\kappa(\Sigma)$ is its condition number. This definition reflects both the magnitude of the regression signal β and the complexity induced by dependence in the latent process.

This SNR is well-defined whenever the eigenvalues of Σ are bounded away from zero and infinity, ensuring that both $\lambda_{\max}(\Sigma)$ and $\kappa(\Sigma)$ are finite. Such conditions hold in many dependent-data settings of interest. In the spatial setting, [Zhan and Datta \(2024\)](#) establishes bounded minimum and maximum eigenvalues for covariance matrices arising from the Matérn family under growing-domain asymptotics, which in turn implies a finite condition number. In the time-series setting, if the latent process is weakly stationary, then its covariance matrix is Toeplitz, since its entries depend only on temporal lag. Classical Toeplitz theory then links the eigenvalues of such covariance matrices to the spectral density, implying that bounded spectral densities yield finite and uniformly controlled eigenvalues; see, for example, [Gray \(2006\)](#) and [Xiao and Wu \(2012\)](#).

In the special case $\Sigma = \tau^2 \mathbf{I}_n$, corresponding to iid errors with no latent-domain dependence, we have $\lambda_{\max}(\Sigma) = \tau^2$ and $\kappa(\Sigma) = 1$, so that SNR reduces to β^2/τ^2 , recovering the classical definition used in iid shuffled regression settings ([Pananjady](#)

et al., 2017; Slawski and Ben-David, 2019). Thus, our definition extends the standard SNR to dependent-data settings.

Rates for Recovery. Here we study recovery guarantees for maximum likelihood estimation under the block permutation model. The most general inferential objective is joint estimation of the block diagonal components of the permutation matrices $(\boldsymbol{\pi}_1, \boldsymbol{\pi}_2)$ in the Cartesian product space $\mathcal{P}_K \times \mathcal{P}_K$ where $\mathcal{P}_K$ denotes the space of all K -dimensional permutation matrices containing $K!$ elements in it, along with the regression parameter β . Under Gaussian assumptions, minimizing the loss $\mathcal{L}(\boldsymbol{\Theta})$ defined in Equation (2) yields the maximum likelihood estimator (MLE). Notice that $\mathcal{L}(\boldsymbol{\Theta})$ can be factored into two parts $\mathcal{L}(\boldsymbol{\Theta}) = \mathcal{L}_1(\boldsymbol{\Theta}) + \mathcal{L}_2(\boldsymbol{\Theta})$, where $\mathcal{L}_1(\boldsymbol{\Theta}) = \left\| \mathbf{P}_{\hat{\boldsymbol{\Pi}}_1, X}^\perp \tilde{\mathbf{Y}}_{\boldsymbol{\Pi}_2} \right\|_2^2$ which governs permutation alignment and $\mathcal{L}_2(\boldsymbol{\Theta}) = \left\| \mathbf{P}_{\boldsymbol{\Pi}_1, X} \tilde{\mathbf{Y}}_{\boldsymbol{\Pi}_2} - \tilde{\mathbf{X}}_{\boldsymbol{\Pi}_1} \beta \right\|_2^2$ governs regression estimation. Assuming $\boldsymbol{\Sigma}$ to be known, for a given value of $(\hat{\boldsymbol{\pi}}_1, \hat{\boldsymbol{\pi}}_2)$ obtained by optimizing the loss function $\mathcal{L}_1$, $\mathcal{L}_2$ can be exactly set to 0, which will give us the closed form estimate of β given by $\hat{\beta} = \left(\tilde{\mathbf{X}}_{\hat{\boldsymbol{\Pi}}_1}^\top \tilde{\mathbf{X}}_{\hat{\boldsymbol{\Pi}}_1} \right)^{-1} \tilde{\mathbf{X}}_{\hat{\boldsymbol{\Pi}}_1}^\top \tilde{\mathbf{Y}}_{\hat{\boldsymbol{\Pi}}_2}$, where $\hat{\boldsymbol{\Pi}}_u = \text{bdiag}(\hat{\boldsymbol{\pi}}_u)$ for $u = 1, 2$. We first characterize conditions under which the permutation matrices are exactly recoverable which will directly imply β is recoverable.

Theorem 1 For $\text{SNR} = \Omega(K^\alpha)$ with $\alpha > 1$ and $B \geq B_*(K, \alpha) := \frac{\alpha \log K}{K^{\alpha-K}}$, with the MLE estimator $\left(\hat{\boldsymbol{\Pi}}_{1,ML}, \hat{\boldsymbol{\Pi}}_{2,ML} \right) = \arg \max_{(\boldsymbol{\pi}_1, \boldsymbol{\pi}_2) \in \mathcal{P}_K \times \mathcal{P}_K} \left\| \mathbf{P}_{\boldsymbol{\Pi}_1, X}^\perp \tilde{\mathbf{Y}}_{\boldsymbol{\Pi}_2} \right\|_2^2$, we have

$$\mathbb{P} \left(\left(\hat{\boldsymbol{\Pi}}_{1,ML}, \hat{\boldsymbol{\Pi}}_{2,ML} \right) \neq (\boldsymbol{\Pi}_1, \boldsymbol{\Pi}_2) \right) = \mathcal{O} \left(K^4 \text{SNR}^{-c_1^*} e^{-2c_1^* B} \right) + \mathcal{O} \left(K^2 e^{-c_2^* B} \right),$$

where the total sample size $n = KB$ for some universal constant $c_1^*, c_2^* > 0$.

The proof is provided in Appendix 2.2. The theorem establishes that under sufficiently strong signal-to-noise conditions that grow polynomially in the block

size K , the permutation matrices can be recovered with exponentially decaying error probability in the number of blocks B . Throughout our analysis, the block size K is treated as fixed, while the number of blocks B is allowed to grow, so that the total sample size $n = KB$ increases through B . In this regime, the lower bound on the required SNR depends only on K and does not scale with the total sample size n . This contrasts with the fully unstructured permutation setting studied in [Pananjady et al. \(2017\)](#), where the SNR requirement scales with n^α . By restricting permutations to local neighborhoods of fixed size, the block structure yields substantially weaker signal requirements while still allowing consistent recovery as $B \rightarrow \infty$.

However, exact recovery of permutations is not necessary for consistent estimation of the regression parameter. The next result establishes that β can be consistently estimated under strictly weaker conditions.

Theorem 2 *Assume $\log \text{SNR} > 1$ and let the number of blocks satisfy $B = \Omega((1+\gamma) \log K / \phi(\text{SNR}))$ for some $\gamma > 0$. Then, for $\hat{\beta} = \left(\widetilde{\mathbf{X}}_{\hat{\Pi}_1}^\top \widetilde{\mathbf{X}}_{\hat{\Pi}_1} \right)^{-1} \widetilde{\mathbf{X}}_{\hat{\Pi}_1}^\top \widetilde{\mathbf{Y}}_{\hat{\Pi}_2}$, where $\left(\hat{\Pi}_1, \hat{\Pi}_2 \right) = \arg \max_{(\pi_1, \pi_2) \in \mathcal{P}_K \times \mathcal{P}_K} \left\| \mathbf{P}_{\hat{\Pi}_1, X}^\perp \widetilde{\mathbf{Y}}_{\hat{\Pi}_2} \right\|_2^2$, and for some $\delta < 1$, we have:*

$$\left| \hat{\beta} - \beta \right| \leq \sqrt{5 \lambda_{\max}(\boldsymbol{\Sigma}) \kappa(\boldsymbol{\Sigma})} \frac{n^{-(1-\delta)/2}}{1 - n^{-(1-\delta)/2}}$$

with probability at least $p = 1 - c_1 K^2 e^{-2B\phi(\text{SNR})} - 2 \exp(-n^\delta)$, where $n = KB$ and $\phi(\text{SNR}) := \frac{(\log \text{SNR} - 1)^2}{\log \text{SNR}}$ and $c_1 > 0$ is a universal constant.

The proof is given in [Appendix 2.3](#). This theorem shows that it is possible to consistently estimate the effect size β , without necessarily correctly estimating the permutation matrices. This provides a motivation for masking or anonymization procedures that preserve population-level association between sensitive variables without requiring individual-level alignment.

Limitation of MLE. Despite the theoretical guarantees established above, directly solving the maximum likelihood problem is computationally infeasible in practice. The optimization over $(\mathbf{\Pi}_1, \mathbf{\Pi}_2)$ requires searching over discrete permutation spaces and can be formulated as a variant of the Quadratic Assignment Problem (QAP), which is known to be NP-hard. Consequently, exact optimization becomes computationally prohibitive even for moderately sized problems. In particular, under the block restriction described earlier, the number of possible permutation pairs grows as $(K!)^2$ within each block, leading to an exponential increase in the search space as K increases.

Beyond the combinatorial complexity, the likelihood surface itself poses additional challenges. The objective function is highly nonconvex due to the interaction between the permutation matrices and the covariance structure $\mathbf{\Sigma}$. In particular, the permutations enter the likelihood through quadratic forms involving $\mathbf{\Sigma}^{-1}$, which couples the effects of $(\mathbf{\Pi}_1, \mathbf{\Pi}_2)$ in a nonlinear manner. As a result, naive optimization strategies such as greedy search or local swap-based updates can easily become trapped in suboptimal configurations and may exhibit substantial sensitivity to initialization. These issues make direct likelihood-based estimation unreliable and computationally unstable for moderate-to-large problem sizes.

To address these challenges, we adopt a scalable approximation strategy. Rather than attempting to solve the discrete optimization problem directly, we introduce a Bayesian formulation in which the unknown permutation matrices and model parameters are treated as latent random variables. This perspective allows us to replace the combinatorial search with inference over a continuous relaxation of the permutation space. In the next section, we develop a variational inference framework that approximates the posterior distribution of the parameters

while remaining computationally tractable. The resulting algorithm provides a practical approximation to the maximum likelihood solution and scales efficiently to moderate-to-large samples.

4. Variational Inference for Joint Permutation and Parameter Estimation

In this section we present our **RE**gression with **P**ermutation **A**lignment via Variational **I**nfe**R**ence (REPAIR) method for parameter estimation. Because direct maximization of the joint likelihood is computationally infeasible due to the combinatorial structure of the permutation matrices, we adopt a Bayesian formulation and approximate the posterior using Variational Bayes. We follow the DGP in Section 2, retaining the block structure in which the permutations are partitioned into B blocks of size K . Under this structure, the per-block DGP is $\mathbf{Y}_i = \boldsymbol{\pi}_X \mathbf{X}_i \beta + \boldsymbol{\pi}_S \mathbf{W}_i + \boldsymbol{\epsilon}_i$ for $i = 1, \dots, B$, where for $\mathbf{S}_i = (s_{i1}, \dots, s_{iK})$ indicating the latent domain indices for block i , $\mathbf{Y}_i$, $\mathbf{X}_i$, $\mathbf{W}_i$, and $\boldsymbol{\epsilon}_i$ denote the corresponding outcome, exposure, latent-domain residual, and iid error vectors, respectively. Define the stacked outcome vector $\mathbb{Y}_n = (\mathbf{Y}_1^\top, \dots, \mathbf{Y}_B^\top)^\top$, and analogously $\mathbb{X}_n$ and $\mathbb{W}_n$. We place a GP prior on $\mathbb{W}_n$ with zero mean and isotropic covariance kernel parameterized by $\boldsymbol{\Gamma} = (\sigma^2, \phi)$, where ϕ controls the range of the correlation between two points in the domain and σ^2 controls the common variance for each component of the latent process. Under this specification, the conditional likelihood (given $\mathbb{X}_n$) for the parameter vector $\boldsymbol{\theta} := (\boldsymbol{\pi}_X, \boldsymbol{\pi}_S, \beta, \sigma^2, \tau^2, \phi, \mathbb{W}_n)^\top$ can be written as

$$L(\mathbb{Y}_n \mid \boldsymbol{\theta}, \mathbb{X}_n) = (2\pi\tau^2)^{-\kappa B/2} \prod_{i=1}^B \exp \left\{ -\frac{1}{2\tau^2} (\mathbf{Y}_i - \boldsymbol{\pi}_X \mathbf{X}_i \beta - \boldsymbol{\pi}_S \mathbf{W}_i)^\top (\mathbf{Y}_i - \boldsymbol{\pi}_X \mathbf{X}_i \beta - \boldsymbol{\pi}_S \mathbf{W}_i) \right\}.$$

Assuming independent priors across the components of $\boldsymbol{\theta}$, the joint prior distribution can be factorized as $p(\boldsymbol{\theta}) := p(\mathbb{W}_n \mid \sigma^2, \phi) \cdot p(\beta) \cdot p(\sigma^2) \cdot p(\phi) \cdot p(\tau^2) \cdot p(\boldsymbol{\pi}_X) \cdot p(\boldsymbol{\pi}_S)$. Since direct prior specification over the discrete permutation space $\mathcal{P}_K$ is computationally intractable ($K!$ configurations per block), we adopt a continuous relaxation following

Maddison et al. (2016); Linderman et al. (2018), placing independent coordinate-wise Gaussian mixture priors on the entries of $\boldsymbol{\pi}_X$ and $\boldsymbol{\pi}_S$, with components centered at 0 and 1 to encourage near-binary matrices.

Based on this construction, we impose the following priors on the model parameters for suitable choices of hyperparameters. Let $\boldsymbol{\Sigma}_n^* = \sigma^2 \mathbf{R}(\phi)$ denote the covariance matrix of the dependent process vector $\mathbb{W}_n$ defined over the $n = KB$ domain points using the isotropic kernel with parameters $\boldsymbol{\Gamma}$ introduced earlier, where $\mathbf{R}(\phi)$ is the corresponding correlation matrix. Denoting $((\boldsymbol{\pi}_X))_{mk} := x_{mk}$ and $((\boldsymbol{\pi}_S))_{mk} := s_{mk}$, the individual prior distributions are given by

$$\begin{aligned} \mathbb{W}_n \mid \sigma^2, \phi &\sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{R}(\phi)), & \beta &\sim \mathcal{N}(0, \sigma_\beta^2), \\ \sigma^2 &\sim \text{Inv-Gamma}(a_1, b_1), & \tau^2 &\sim \text{Inv-Gamma}(a_2, b_2), \\ \phi &\sim \text{Unif}(0, \sqrt{2}), & x_{mk} &\sim \tfrac{1}{2} \mathcal{N}(0, \eta_x^2) + \tfrac{1}{2} \mathcal{N}(1, \eta_x^2), \\ s_{mk} &\sim \tfrac{1}{2} \mathcal{N}(0, \eta_s^2) + \tfrac{1}{2} \mathcal{N}(1, \eta_s^2). \end{aligned}$$

Let $\boldsymbol{\eta}$ denote the vector of hyperparameters used in the prior distribution of $\boldsymbol{\theta} = (\mathbb{W}_n, \beta, \sigma^2, \phi, \tau^2, \boldsymbol{\pi}_X, \boldsymbol{\pi}_S)^\top$. We approximate the posterior distribution of the parameter vector $\boldsymbol{\theta}$ using variational inference over a tractable family of distributions $q(\boldsymbol{\theta} \mid \boldsymbol{\lambda})$, where $\boldsymbol{\lambda}$ denotes the parameters governing the joint variational density of $\boldsymbol{\theta}$ conditional on $\mathbb{X}_n$. The approximate posterior is obtained by optimizing $\boldsymbol{\lambda}$ to minimize the Kullback-Leibler (KL) divergence between the variational density $q(\cdot \mid \boldsymbol{\lambda})$ and the true posterior conditional on $\mathbb{X}_n$. Equivalently, this corresponds to maximizing the Evidence Lower BOund (ELBO):

$$\mathcal{L}(\boldsymbol{\lambda}) := \mathbb{E}_{q(\cdot \mid \boldsymbol{\lambda})} [\log L(\mathbb{Y}_n, \boldsymbol{\theta} \mid \mathbb{X}_n) - \log q(\boldsymbol{\theta} \mid \boldsymbol{\lambda}, \mathbb{X}_n)]. \quad (3)$$

We assume a mean-field variational approximation for the target posterior distribution of $\boldsymbol{\theta}$. Under this factorization, the conjugacy results of [Ren et al. \(2011\)](#) for exponential-family models yield closed-form expressions for most of the variational densities corresponding to the model parameters. For parameters whose variational densities do not admit closed-form solutions, the corresponding variational densities are estimated by directly maximizing the ELBO.

Based on the prior structure and the assumed likelihood model, Gaussian variational families arise for the dependent random effect vector $\mathbb{W}_n$, parameterized by $(\boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W)$, and for β , parameterized by $(\mu_{\lambda,\beta}, \sigma_{\lambda,\beta}^2)$. The variance parameters σ^2 and τ^2 follow inverse-gamma variational families parameterized by $(\lambda_{a_1}, \lambda_{b_1})$ and $(\lambda_{a_2}, \lambda_{b_2})$, respectively. The scale parameter ϕ is associated with a more complicated variational density whose form, along with closed form solution of the parameters defining the above mentioned variational distributions are derived in [Appendix 1.1](#).

For the permutation matrices, we adopt the continuous relaxation variational family of [Linderman et al. \(2018\)](#). Given parameters $\zeta := (\mathbf{M}, \mathbf{V})$, a doubly stochastic matrix $\widetilde{\mathbf{M}}$ is obtained by applying the Sinkhorn–Knopp algorithm ([Sinkhorn and Knopp, 1967](#)) to $\mathbf{M}$, then a perturbed matrix $\boldsymbol{\Psi} = \widetilde{\mathbf{M}} + \mathbf{V} \odot \mathbf{Z}$ (with $\mathbf{Z}$ standard normal) is constructed. A temperature-controlled interpolation $\boldsymbol{\pi}^* = \kappa \boldsymbol{\Psi} + (1 - \kappa) \text{round}(\boldsymbol{\Psi})$ produces a relaxed permutation, where $\text{round}(\cdot)$ applies the Hungarian algorithm ([Kuhn, 1955](#)) to get the closest permutation matrix and $\kappa \rightarrow 0$ recovers discrete permutations. Letting $\mathbf{Z} = g_\kappa^{-1}(\boldsymbol{\pi}^*; \zeta)$ denote the inverse transformation parametrized by ζ , the induced variational density is

$$q_\kappa(\boldsymbol{\pi}^* \mid \zeta) = \prod_{m=1}^K \prod_{k=1}^K \frac{1}{\kappa v_{mk}} \mathcal{N}(z_{mk}; 0, 1) \mathbb{I}\{\boldsymbol{\pi}^* \in \mathcal{G}_\kappa\}, \quad (4)$$

where $z_{mk} = [g_\kappa^{-1}(\boldsymbol{\pi}^*; \zeta)]_{mk}$, v_{mk} denotes the (m, k) -th entry of $\mathbf{V}$, $\mathcal{G}_\kappa$ denotes the image of $g_\kappa(\cdot \mid \zeta)$, and the products run over the $K \times K$ within-block entries. For the two permutation matrices $\boldsymbol{\pi}_X$ and $\boldsymbol{\pi}_S$, we assign the above variational density as $q_{\kappa_X}(\cdot \mid \zeta_X)$ and $q_{\kappa_S}(\cdot \mid \zeta_S)$, respectively, where $\zeta_Q = (\mathbf{M}_Q, \mathbf{V}_Q)$ for $Q \in \{X, S\}$.

Hence $\boldsymbol{\lambda} = (\mu_{\lambda, \beta}, \sigma_{\lambda, \beta}^2, \boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W, \lambda_{a_1}, \lambda_{b_1}, \lambda_{a_2}, \lambda_{b_2}, \zeta_X, \zeta_S)^\top$ denote the vector parameterizing the joint variational density of $\boldsymbol{\theta}$. Under the mean-field assumption, the joint variational density decomposes as

$$\begin{aligned} q(\boldsymbol{\theta} \mid \boldsymbol{\lambda}) &:= q_W(\mathbb{W}_n \mid \boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W) \cdot q_\beta(\beta \mid \mu_{\lambda, \beta}, \sigma_{\lambda, \beta}^2) \cdot q_{\sigma^2}(\sigma^2 \mid \lambda_{a_1}, \lambda_{b_1}) \cdot q_{\tau^2}(\tau^2 \mid \lambda_{a_2}, \lambda_{b_2}) \\ &\cdot q_\phi(\phi) \cdot q_{\kappa_X}(\boldsymbol{\pi}_X \mid \zeta_X) \cdot q_{\kappa_S}(\boldsymbol{\pi}_S \mid \zeta_S). \end{aligned} \tag{5}$$

Except for the parameters associated with the permutation matrices, closed-form solutions are available for all other variational parameters. The variational parameters corresponding to the permutation matrices do not admit closed-form updates and are therefore estimated via numerical optimization that maximizes the ELBO with respect to them. All the details are provided in Appendix 1.1.

In addition to the parameters $\boldsymbol{\lambda}$, several auxiliary quantities arise in computing the closed-form updates and evaluating the ELBO for the permutation matrices. One such quantity is $\mathbb{E}_{q_\phi}[\mathbf{R}(\phi)^{-1}]$, where the expectation is taken with respect to the variational density of ϕ . Another set of quantities are $\boldsymbol{\lambda}_{\boldsymbol{\pi}_X} = (\mathbf{M}_X^*, \mathbf{V}_X^*)$ and $\boldsymbol{\lambda}_{\boldsymbol{\pi}_S} = (\mathbf{M}_S^*, \mathbf{V}_S^*)$, where $\mathbf{M}_X^* := \mathbb{E}_{q_{\kappa_X}}[\boldsymbol{\pi}_X]$, $\mathbf{V}_X^* := \mathbb{E}_{q_{\kappa_X}}[\boldsymbol{\pi}_X^\top \boldsymbol{\pi}_X]$, and similarly for $\boldsymbol{\pi}_S$. These expectations appear in multiple steps of the parameter updates and must therefore be estimated as part of the optimization procedure, and thus we include them within the parameter vector $\boldsymbol{\lambda}$ defined earlier.

Since each variational parameter update depends on the current values of the others, we resort to an iterative estimation procedure in which each parameter in

$\boldsymbol{\lambda}$, together with the auxiliary quantities described above, is updated using the most recent values of the remaining parameters. After each outer iteration, the global ELBO (Appendix 1.2) is evaluated; the procedure terminates when the ELBO improvement falls below a threshold ϵ . The temperatures κ_X and κ_S are annealed exponentially during optimization. Algorithm 1 summarizes the procedure.

Algorithm 2 describes the procedure used to recover the permutation matrices from the estimated variational parameters. The matrices $\widehat{\mathbf{M}}_X^*$ and $\widehat{\mathbf{M}}_S^*$, which correspond to the posterior expectations of $\boldsymbol{\pi}_X$ and $\boldsymbol{\pi}_S$, may not themselves be valid permutation matrices. Therefore, we first project these matrices onto the Birkhoff polytope $\mathcal{B}_K$, the space of all doubly stochastic matrices. This projection is performed using the Sinkhorn–Knopp algorithm. Once a doubly stochastic approximation is obtained, we recover the closest permutation matrix by solving a linear assignment problem using the Hungarian algorithm. The same procedure is applied to both $\widehat{\mathbf{M}}_X^*$ and $\widehat{\mathbf{M}}_S^*$ to obtain the final estimates $\widehat{\boldsymbol{\pi}}_X$ and $\widehat{\boldsymbol{\pi}}_S$. The following section evaluates the finite-sample performance of REPAIR.

5. Simulation Studies

We assess REPAIR’s finite-sample performance across varying block geometries and SNR levels, comparing it to two benchmarks: **FullGP**, an oracle method with true permutations known, fitted by MLE; and **ArealGP**, which aggregates exposure and outcome to the block level and applies GLS. All methods use a correctly-specified exponential covariance kernel. The design crosses $B \in \{49, 81, 100, 121\}$ blocks and $K \in \{6, 8, 10, 12, 20\}$ observations per block ($n = KB$ from 294 to 2420) with two signal levels $\beta \in \{2, 8\}$ controlling the levels of SNR values. Exposure is drawn as $X_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$; the spatial random effect follows $\mathbf{W} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{R}(\phi))$ with an exponential kernel ($\sigma^2 = 5$, $\phi = 0.5$); the iid noise has variance $\tau^2 = 0.5$; and

Algorithm 1 VB Algorithm to estimate the parameters of the variational distribution of θ

Require: the hyperparameters η , learning rates (l_X, l_S) , threshold ϵ for the ELBO, initial values of the variational density parameters $\lambda^{(0)}$.

Ensure: Variational parameter estimates:

$$\lambda^{(T)} = \left\{ \mu_{\lambda, \beta}^{(T)}, \sigma_{\lambda, \beta}^{2(T)}, \mu_W^{(T)}, \Sigma_W^{(T)}, \lambda_{a_1}^{(T)}, \lambda_{b_1}^{(T)}, \lambda_{a_2}^{(T)}, \lambda_{b_2}^{(T)}, \mathbb{E}^{(T)}[\mathbf{R}(\phi)^{-1}], \zeta_X^{(T)}, \zeta_S^{(T)}, \mathbf{M}_X^{*(T)}, \mathbf{V}_X^{*(T)}, \mathbf{M}_S^{*(T)}, \mathbf{V}_S^{*(T)} \right\}$$

where $T := T(\epsilon)$.

while $\text{ELBO}^{(t+1)} - \text{ELBO}^{(t)} > \epsilon$ **do**

Step 1: Update the distribution of $\beta \sim \mathcal{N}(\mu_{\lambda, \beta}^{(t)}, \sigma_{\lambda, \beta}^{2(t)})$ where

$$\sigma_{\lambda, \beta}^{2(t)} = \left(\frac{\lambda_{a_2}^{(t-1)}}{\lambda_{b_2}^{(t-1)}} \sum_{i=1}^B \mathbf{X}_i^\top \mathbf{V}_X^{*(t-1)} \mathbf{X}_i + \frac{1}{\sigma_{\beta}^{2(t-1)}} \right)^{-1} \text{ and}$$

$$\mu_{\lambda, \beta}^{(t)} = \sigma_{\lambda, \beta}^{2(t)} \left(\frac{\lambda_{a_2}^{(t-1)}}{\lambda_{b_2}^{(t-1)}} \sum_{i=1}^B \mathbf{X}_i^\top \mathbf{M}_X^{*(t-1)^\top} \left(\mathbf{Y}_i - \mathbf{M}_S^{*(t-1)} \mu_{W_i}^{(t-1)} \right) \right).$$

Step 2: Update the distribution of $\mathbb{W}_n \sim \mathcal{N}(\mu_W^{(t)}, \Sigma_W^{(t)})$ where

$$\Sigma_W^{(t)} = \left(\frac{\lambda_{a_2}^{(t-1)}}{\lambda_{b_2}^{(t-1)}} \text{diag}(\mathbf{V}_S^{*(t-1)}) + \frac{\lambda_{a_1}^{(t-1)}}{\lambda_{b_1}^{(t-1)}} \mathbb{E}^{(t-1)}[\mathbf{R}(\phi)^{-1}] \right)^{-1} \text{ and}$$

$$\mu_W^{(t)} = \Sigma_W^{(t)} \left(\frac{\lambda_{a_2}^{(t-1)}}{\lambda_{b_2}^{(t-1)}} \text{diag}(\mathbf{M}_S^{*(t-1)})^\top \left(\mathbb{Y}_n - \mu_{\lambda, \beta}^{(t)} \text{diag}(\mathbf{M}_X^{*(t-1)}) \mathbb{X}_B \right) \right)$$

Step 3: Update the distribution of $\sigma^2 \sim \text{Inv} - \text{Gamma}(\lambda_{a_1}^{(t)}, \lambda_{b_1}^{(t)})$ where

$$\lambda_{a_1}^{(t)} = \frac{KB}{2} + a_1 \text{ and } \lambda_{b_1}^{(t)} = \frac{1}{2} \left[\text{tr} \left(\mathbb{E}^{(t-1)}[\mathbf{R}(\phi)^{-1}] \Sigma_W^{(t)} \right) + \mu_W^{(t)\top} \mathbb{E}^{(t-1)}[\mathbf{R}(\phi)^{-1}] \mu_W^{(t)} \right] + b_1$$

Step 4: Update the distribution of $\tau^2 \sim \text{Inv} - \text{Gamma}(\lambda_{a_2}^{(t)}, \lambda_{b_2}^{(t)})$ where

$$\lambda_{a_2}^{(t)} = \frac{KB}{2} + a_2 \text{ and}$$

$$\lambda_{b_2}^{(t)} = \sum_{i=1}^B \left(\mathbf{Y}_i - \mu_{\lambda, \beta}^{(t)} \mathbf{M}_X^{*(t-1)} \mathbf{X}_i - \mathbf{M}_S^{*(t-1)} \mu_{W_i}^{(t)} \right)^\top \left(\mathbf{Y}_i - \mu_{\lambda, \beta}^{(t)} \mathbf{M}_X^{*(t-1)} \mathbf{X}_i - \mathbf{M}_S^{*(t-1)} \mu_{W_i}^{(t)} \right)$$

$$+ \mu_{\lambda, \beta}^{2(t)} \sum_{i=1}^B \mathbf{X}_i^\top \left(\mathbf{V}_X^{*(t-1)} - \mathbf{M}_X^{*(t-1)^\top} \mathbf{M}_X^{*(t-1)} \right) \mathbf{X}_i + \sigma_{\lambda, \beta}^{2(t)} \sum_{i=1}^B \mathbf{X}_i^\top \mathbf{V}_X^{*(t-1)} \mathbf{X}_i$$

$$+ \sum_{i=1}^B \left[\mu_{W_i}^{(t)\top} \left(\mathbf{V}_S^{*(t-1)} - \mathbf{M}_S^{*(t-1)^\top} \mathbf{M}_S^{*(t-1)} \right) \mu_{W_i}^{(t)} + \text{tr} \left(\mathbf{V}_S^{*(t-1)} \Sigma_{W_i}^{(t)} \right) \right] + b_2$$

Step 5: Update the distribution on ϕ which is proportional to

$$|\mathbf{R}(\phi)|^{-\frac{1}{2}} \exp \left(-\frac{\lambda_{a_1}^{(t)}}{2\lambda_{b_1}^{(t)}} \left[\text{tr} \left(\mathbf{R}(\phi)^{-1} \Sigma_W^{(t)} \right) + \mu_W^{(t)\top} \mathbf{R}(\phi)^{-1} \mu_W^{(t)} \right] \right) \text{ and compute}$$

$$\mathbb{E}_{q_\phi}^{(t)}[\mathbf{R}(\phi)^{-1}] \text{ using importance sampling.}$$

Step 6: Using the last 5 steps and the learning rates l_X and l_S , minimize the ELBO

$$\text{for } \pi_X \text{ and } \pi_S \text{ to get the parameters } \zeta_X^{(t)}, \zeta_S^{(t)}, \mathbf{M}_X^{*(t)}, \mathbf{V}_X^{*(t)}, \mathbf{M}_S^{*(t)}, \mathbf{V}_S^{*(t)}.$$

end while

Algorithm 2 Algorithm for estimating the permutation matrices

Require: The estimated parameters $\widehat{\mathbf{M}}_X^* = \widehat{\mathbb{E}}_{q_{\kappa_X}}[\pi_X]$ and $\widehat{\mathbf{M}}_S^* = \widehat{\mathbb{E}}_{q_{\kappa_S}}[\pi_S]$

Ensure: $\widehat{\pi}_X, \widehat{\pi}_S$

Step 1: Project $\widehat{\mathbf{M}}_X^*$ onto Birkhoff Polytope $\mathcal{B}_K$ to get doubly stochastic $\widehat{\mathcal{B}}_X$.

Step 2: Estimate the nearest permutation matrix $\widehat{\pi}_X$ from $\widehat{\mathcal{B}}_X$.

Step 3: Repeat Steps 1–2 for $\widehat{\mathbf{M}}_S^*$ to obtain $\widehat{\pi}_S$.

within-block Hamming distances $d_H(\boldsymbol{\pi}_X, \mathbf{I}_K)$, $d_H(\boldsymbol{\pi}_S, \mathbf{I}_K)$ are drawn uniformly from $\{1, \dots, K\}$ and held fixed across 100 replicates. Full specifications, including the grid construction procedure and hyperparameter settings for REPAIR, are provided in Supplement Section 4.

Simulation Results. We focus on recovery of β and $(\boldsymbol{\pi}_X, \boldsymbol{\pi}_S)$; variance and covariance parameter estimates are reported in Figure 1 of the Supplement.

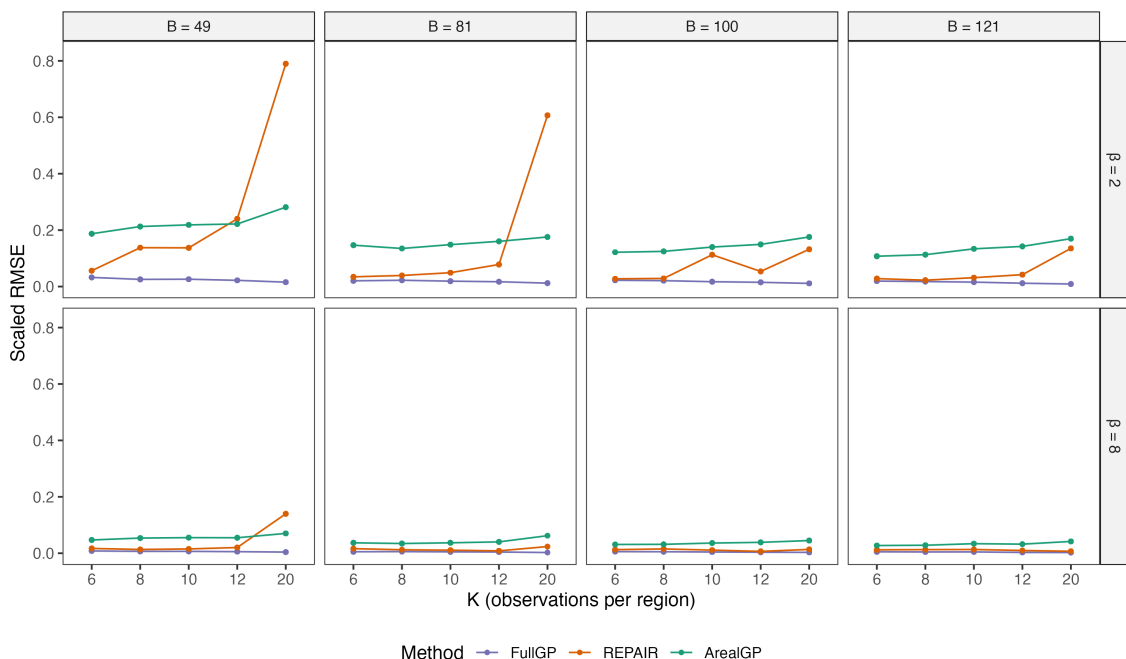


Fig. 2: Scaled RMSE of $\hat{\beta}$ across simulation settings. Columns correspond to the number of regions B and the horizontal axis shows the number of observations per region K . Results are reported for two SNR regimes, $\beta = 2$ and $\beta = 8$, comparing FullGP, ArealGP, and REPAIR.

Figure 2 summarizes $\hat{\beta}$ estimation accuracy. REPAIR consistently outperforms ArealGP, with the gap widening as B increases. RMSE for REPAIR rises with K , reflecting the harder permutation problem at larger block sizes, consistent with Theorem 2. Under high SNR ($\beta = 8$), REPAIR is nearly indistinguishable from

the oracle FullGP; under low SNR ($\beta = 2$), the gap widens but REPAIR retains its advantage over ArealGP across most (K, B) regimes. Variance and covariance parameter estimates (Figure 1 in the Supplement) show higher RMSE for REPAIR than for FullGP, consistent with the known limitations of mean-field variational approximations for nuisance covariance parameters.

Figure 2 in the Supplement reports permutation recovery. A consistent pattern is that π_X is recovered with notably higher probability than π_S across all regimes. This asymmetry is interpretable: π_X enters the likelihood directly through the signal term $\Pi_X \mathbf{X} \beta$, so a stronger exposure signal provides more information for identifying it, while π_S affects only the latent process term $\Pi_S \mathbf{W}$ whose recovery is governed by the spatial SNR alone. In several regimes with larger K , recovery of π_S degrades substantially while π_X remains well-identified. Crucially, β estimation remains accurate throughout (Figure 2), providing direct empirical confirmation of the theoretical separation in Theorems 1 and 2: consistent regression recovery does not require consistent permutation recovery.

6. Real Data Analysis

We illustrate REPAIR using the Meuse soil dataset (Pebesma et al., 2012), recording heavy metal concentrations and covariates at 155 locations on the river Meuse floodplain, Netherlands. We take $Y = \log(1 + \text{zinc})$ as the outcome and elevation as the exposure, center both variables, and rescale spatial coordinates to the unit square. Five observations are removed to give $n = 150$; locations are ordered by the first spatial coordinate and partitioned into contiguous blocks with exposure values and coordinates permuted within each block to induce the doubly-unlinked structure. We consider two schemes: (a) 15 blocks of size 10 and (b) 30 blocks of size

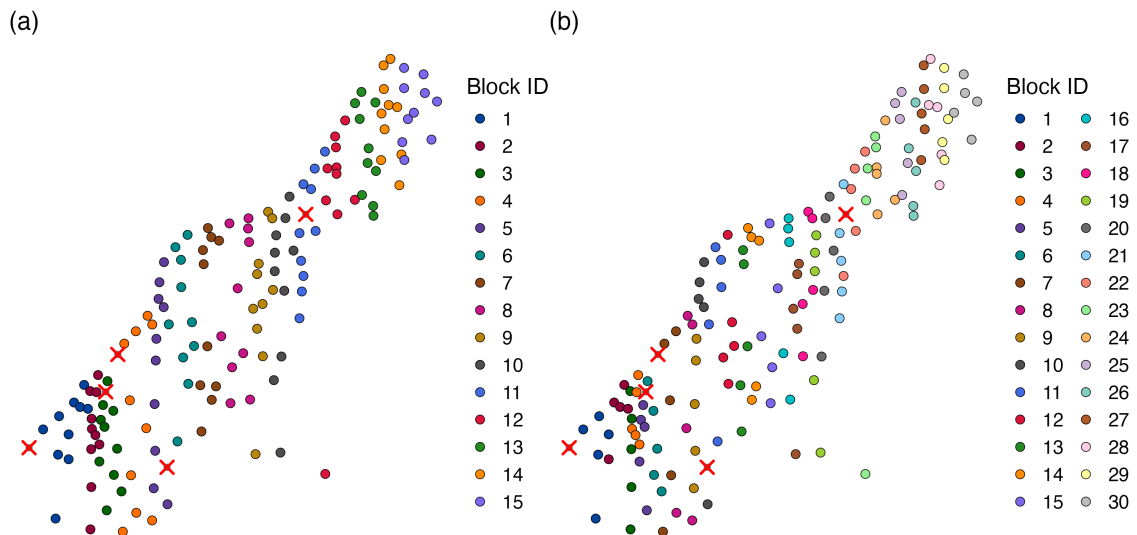


Fig. 3: Partitions for the Meuse analysis after dropping five observations: (a) 15×10 and (b) 30×5 . Colored points denote block memberships, and red crosses indicate the removed observations.

5 (Figure 3). Full dataset background and preprocessing details are in Supplement Section 5.

We compare **ArealGP**, **REPAIR**, and the oracle **FullGP** in effect size estimation. All three methods recover a negative effect of elevation on zinc concentration across both blocking schemes, consistent with the known pattern of higher zinc concentrations near the low-lying riverbank. The estimated regression coefficients are: -0.2926 (FullGP, both partitions), and under the 15×10 and 30×5 partitions respectively, -0.2317 and -0.3965 for ArealGP, and -0.2102 and -0.2831 for REPAIR.

Under the finer 30×5 partition, REPAIR (-0.2831) is close to the FullGP benchmark (-0.2926) and outperforms ArealGP (-0.3965), which overshoots substantially. Under the coarser 15×10 partition, where within-block permutation

recovery is only partial, REPAIR (-0.2102) is farther from the benchmark than ArealGP (-0.2317); both estimates preserve the correct sign and order of magnitude. This ordering is consistent with Theorem 2: when permutation recovery degrades (large K , few blocks), REPAIR’s advantage over a naive aggregator diminishes, yet the regression signal remains estimable.

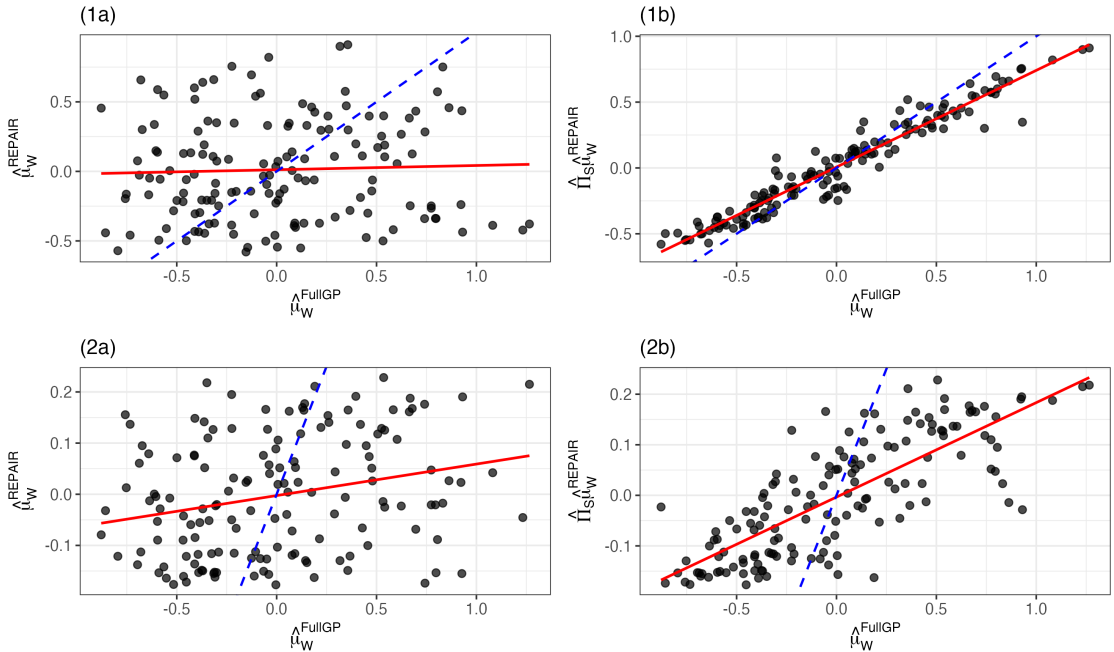


Fig. 4: Comparison of latent surface estimates ($\hat{\mu}_W$, the variational posterior mean for $\mathbb{W}_n$) from REPAIR and FullGP under the two blocking schemes. Panels (1a)–(2a) plot $\hat{\mu}_W^{\text{REPAIR}}$ against $\hat{\mu}_W^{\text{FullGP}}$; panels (1b)–(2b) plot the permutation-corrected $\hat{\Pi}_S \hat{\mu}_W^{\text{REPAIR}}$ against $\hat{\mu}_W^{\text{FullGP}}$. Dashed blue line: 45° reference; solid red line: least-squares fit.

Figure 4 reinforces these conclusions at the level of the latent spatial surface. In the 30×5 case, $\hat{\Pi}_S \hat{\mu}_W^{\text{REPAIR}}$ aligns closely with $\hat{\mu}_W^{\text{FullGP}}$ once the estimated permutation is applied, with points lying near the 45° line. In the 15×10 case the scatter is more diffuse, reflecting partial permutation recovery. Together, the

Meuse results illustrate a practically important property: REPAIR can recover the exposure effect and latent spatial surface accurately when the block structure is sufficiently fine, while even under the coarser partition the direction of the effect and a reasonable magnitude are preserved, consistent with the theoretical separation between regression and permutation recovery conditions.

7. Discussion

The central finding of this paper is that regression recovery and permutation recovery are governed by strictly different signal-to-noise conditions in the doubly-unlinked dependent-data setting. The latent dependence structure provides enough information for consistent estimation of β even when it cannot resolve individual-level correspondences, a separation formalised in Theorems 1 and 2 that has no counterpart in the iid shuffled regression literature (Pananjady et al., 2017). Its origin lies in the covariance structure of the latent process: dependence constrains the effective permutation space in a way that benefits regression estimation without fully identifying the permutations themselves.

A consequence worth drawing out concerns spatial data release. Standard geoprivacy mechanisms (spatial aggregation, noise addition) treat the tension between confidentiality and analytical utility as fundamental, purchasing one at the cost of the other. Our framework suggests a different possibility rooted in the Gaussian process assumption that most spatial analyses already adopt. Under a Gaussian process prior, individual locations are modeled as samples from a smooth latent surface, and the scientifically relevant object is the population-level exposure effect β , not the individual-level spatial attribution encoded by Π_S . The question our theory addresses, whether exact spatial correspondence is necessary to recover β , follows directly from this modelling perspective. Theorem 2 establishes

that the answer is no: a structured scrambling of both covariate assignments and spatial coordinates can fall below the threshold sufficient for permutation recovery while leaving β consistently estimable. This opens the possibility of data-release mechanisms that preserve population-level inference while masking individual-level correspondence, a question with broad implications for spatial, temporal, and spatio-temporal applications in environmental epidemiology, digital mental health, and spatial transcriptomics.

Several open problems follow directly from this work. First, establishing minimax lower bounds for the doubly-unlinked problem would precisely characterise the information-theoretic cost of the second broken link and determine whether the upper bounds in Theorem 2 are sharp, completing the characterisation of recovery conditions. Second, the block structure studied here preserves coarse-scale domain ordering while permuting within local neighbourhoods; the complementary regime, which permutes blocks while preserving within-block structure, may better reflect certain anonymisation or distributed data-collection mechanisms, and the behaviour of the SNR separation in this alternative warrants separate analysis. Third, extending the framework to multivariate exposures $\mathbf{X} \in \mathbb{R}^p$ raises the question of whether the separation gap between regression and permutation recovery thresholds narrows with covariate dimension, directly relevant to the privacy interpretation since the gap determines the strength of the anonymisation guarantee. Finally, the graph-indexed case, where dependence is encoded by network structure rather than a covariance kernel, connects the doubly-unlinked regime to the graph matching literature (Arroyo et al., 2021; Lyzinski et al., 2014; Fishkind et al., 2019; Dawn and Arroyo, 2025) and may yield sharper recovery conditions when the graph has known spectral properties.

8. Data availability statement

The data used in this study are openly available in the Comprehensive R Archive Network (CRAN) as part of the `sp` package (Pebesma and Bivand, 2005) at <https://CRAN.R-project.org/package=sp>. The `meuse` dataset can be loaded directly in R via `library(sp); data(meuse)`. The original field data were collected by Rikken and Van Rijn (1993).

9. Software

The REPAIR software and vignette are available at <https://github.com/isayantan/SpatialReg-Unlinked>.

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References

- J. Arroyo, D. L. Sussman, C. E. Priebe, and V. Lyzinski. Maximum likelihood estimation and graph matching in errorfully observed networks. *Journal of Computational and Graphical Statistics*, 30(4):1111–1123, 2021. doi: 10.1080/10618600.2021.1872582.
- E. B. Banning. Seriation. *The Archaeologist's Laboratory: The Analysis of Archaeological Evidence*, pages 309–316, 2020.
- C. Beuthner, J. Breuer, and S. Jünger. Data linking—linking survey data with geospatial, social media, and sensor data. *GESIS Survey Guidelines*, 2021.
- C. A. Cassa, S. C. Wieland, and K. D. Mandl. Re-identification of home addresses from spatial locations anonymized by gaussian skew. *International journal of health geographics*, 7(1):45, 2008.
- T. Dawn and J. Arroyo. Covariate-assisted graph matching. *arXiv preprint arXiv:2512.11761*, 2025.

-
- C. Durot and D. Mukherjee. Minimax optimal rates of convergence in the shuffled regression, unlinked regression, and deconvolution under vanishing noise. *arXiv preprint arXiv:2404.09306*, 2024.
- E. Fabrizi, N. Salvati, and M. Slawski. Accounting for mismatch error in small area estimation with linked data. *Journal of Survey Statistics and Methodology*, 2025. doi: 10.1093/jssam/smaf011. Online ahead of print.
- D. E. Fishkind, S. Adali, H. G. Patsolic, L. Meng, D. Singh, V. Lyzinski, and C. E. Priebe. Seeded graph matching. *Pattern Recognition*, 87:203–215, 2019. doi: 10.1016/j.patcog.2018.09.014.
- R. M. Gray. Toeplitz and circulant matrices: A review. 2006.
- Y. Gu, D. Blaauw, and J. D. Welch. Bayesian inference of rna velocity from multi-lineage single-cell data. *BioRxiv*, pages 2022–07, 2022.
- D. J. Hsu, K. Shi, and X. Sun. Linear regression without correspondence. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. Polynomial-time approximations for certain regimes.
- H. W. Kuhn. The hungarian method for the assignment problem. *Naval research logistics quarterly*, 2(1-2):83–97, 1955.
- P. Lahiri and M. D. Larsen. Regressionanalysis with linked data. *Journal of the American Statistical Association*, 100(469):222–230, 2005.
- Y. Lin. Geo-indistinguishable masking: enhancing privacy protection in spatial point mapping. *Cartography and Geographic Information Science*, 50(6):608–623, 2023.
- S. Linderman, G. Mena, H. Cooper, L. Paninski, and J. Cunningham. Reparameterizing the birkhoff polytope for variational permutation inference. In *International Conference on Artificial Intelligence and Statistics*, pages 1618–1627. PMLR, 2018.
- H. Liu and A. Scaglione. Shuffled linear regression via spectral matching. *IEEE Transactions on Signal Processing*, 2025.
- H. Liu, A. Scaglione, and H.-T. Wai. Blind graph matching using graph signals. *IEEE Transactions on Signal Processing*, 72:1766–1781, 2024.
- E. Loiola, N. Abreu, P. Boaventura-Netto, P. Hahn, and T. Querido. A survey of the quadratic assignment problem. *European Journal of Operational Research*, 176:657–690, 01 2007. doi: 10.1016/j.ejor.2005.09.032.
- V. Lyzinski, D. E. Fishkind, and C. E. Priebe. Seeded graph matching for correlated Erdős–Rényi graphs. *Journal of Machine Learning Research*, 15(1):3513–3540, 2014.
- C. J. Maddison, A. Mnih, and Y. W. Teh. The concrete distribution: A continuous relaxation of discrete random variables. *arXiv preprint arXiv:1611.00712*, 2016.
- M. Nitzan, N. Karaikos, N. Friedman, and N. Rajewsky. Charting a tissue from single-cell transcriptomes. *Nature*, 577:130–137, 2019.
- A. Pananjady, M. J. Wainwright, and T. A. Courtade. Linear regression with shuffled data: Statistical and computational limits of permutation recovery. *IEEE Transactions on Information Theory*, 64(5):3286–3300, 2017.

- A. Pananjady, M. J. Wainwright, and T. A. Courtade. Linear regression with shuffled data: Statistical and computational limits of permutation recovery. *IEEE Transactions on Information Theory*, 64(5):3286–3300, 2018. Shows hardness of ML permutation estimation and statistical thresholds.
- E. Pebesma and R. Bivand. *sp: Classes and Methods for Spatial Data*, 2005. URL <https://CRAN.R-project.org/package=sp>. R package, version 2.1-4.
- E. Pebesma, R. Bivand, M. E. Pebesma, S. RColorBrewer, and A. Collate. Package ‘sp’. *The Comprehensive R Archive Network*, 9, 2012.
- Q. Ren, S. Banerjee, A. O. Finley, and J. S. Hodges. Variational bayesian methods for spatial data analysis. *Computational statistics & data analysis*, 55(12):3197–3217, 2011.
- M. G. J. Rikken and R. P. G. Van Rijn. Soil pollution with heavy metals—an inquiry into spatial variation, cost of mapping and the risk evaluation of copper, cadmium, lead and zinc in the floodplains of the Meuse west of Stein, the Netherlands. Doctoraalveldwerkverslag, Department of Physical Geography, Utrecht University, 1993.
- C. Rossant, S. N. Kadir, D. F. Goodman, J. Schulman, M. L. Hunter, A. B. Saleem, A. Grosmark, M. Belluscio, G. H. Denfield, A. S. Ecker, et al. Spike sorting for large, dense electrode arrays. *Nature neuroscience*, 19(4):634–641, 2016.
- R. Sinkhorn and P. Knopp. Concerning nonnegative matrices and doubly stochastic matrices. *Pacific Journal of Mathematics*, 21(2):343–348, 1967.
- M. Slawski and E. Ben-David. Linear regression with sparsely permuted data. *Electronic Journal of Statistics*, 13(1):1–36, 2019. Models sparse permutation errors and proposes two-stage approaches.
- M. Slawski, B. T. West, P. Bukke, Z. Wang, G. Diao, and E. Ben-David. A general framework for regression with mismatched data based on mixture modeling. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 188(3):896–919, 2024. doi: 10.1093/jrssa/qnae083.
- J. Unnikrishnan, S. Haghighatshoar, and M. Vetterli. Unlabeled sensing: Solving a linear system with unordered measurements. In *2015 53rd Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 786–793. IEEE, 2015.
- Z. Wang, E. Ben-David, and M. Slawski. Estimation in exponential family regression based on linked data contaminated by mismatch error. *Statistics and Its Interface*, 16:379–396, 2023. doi: 10.4310/22-SII726.
- H.-J. Westra, R. C. Jansen, R. S. Fehrmann, G. J. te Meerman, D. Van Heel, C. Wijmenga, and L. Franke. Mixupmapper: correcting sample mix-ups in genome-wide datasets increases power to detect small genetic effects. *Bioinformatics*, 27(15):2104–2111, 2011.
- H. Xiao and W. B. Wu. Covariance matrix estimation for stationary time series. *The Annals of Statistics*, pages 466–493, 2012.
- W. Zhan and A. Datta. Neural networks for geospatial data. *Journal of the American Statistical Association*, pages 1–21, 2024.
- H. Zhang and P. Li. Optimal estimator for unlabeled linear regression. In *International Conference on Machine Learning*, pages 11153–11162. PMLR, 2020.